Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

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III SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END UP SEMESTER EXAMINATIONS, NOVEMBER 2024

ELECTROMAGNETIC FIELD THEORY [ELE 2124]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 29 NOVEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Consider a vector $\bar{\boldsymbol{H}} = \left(\frac{A}{\rho}\right)\boldsymbol{a}_{\emptyset}$, where A is a constant and ρ is the radius of the cylinder. From the fundamental concepts, show the steps involved in transforming the vector $\bar{\boldsymbol{H}}$ to its equivalent in a spherical coordinate system. ***(03)***

1B. Let $\bar{\boldsymbol{D}} = (2y^2 + z)\boldsymbol{a_x} + 4\text{xy}\boldsymbol{a_y} + \text{x}\boldsymbol{a_z}\ C/m^2$. Find: ***(03)***

a) The volume charge density at (-1, 0, 3)

b) The total charge enclosed by the cube defined by $x = 0\ and\ 1; y = 0\ and\ 1; z = 0\ and\ 1$

1C. A unit normal vector from region-2 ($\mu = 3\mu_0$) to region-1 ($\mu = \mu_0$) is defined as $\boldsymbol{a_{n21}} = {}^{(4\boldsymbol{a_x} + 4\boldsymbol{a_y} - 2\boldsymbol{a_z})}/_{6}$. If $\bar{\boldsymbol{H}}_{\boldsymbol{1}} = (\,20\boldsymbol{a_x} + 2\boldsymbol{a_y} + 12\boldsymbol{a_z})$A/m, and $\bar{\boldsymbol{H}}_{\boldsymbol{2}} = (\,H_{2x}\boldsymbol{a_x} - 5\boldsymbol{a_y} + 4\boldsymbol{a_z})$A/m, estimate the following parameters: ***(04)***

a) $'H'_{2x}$

b) surface current density K on the interface.

c) The angles $\bar{\boldsymbol{B}}_{\boldsymbol{1}}$ and $\bar{\boldsymbol{B}}_{\boldsymbol{2}}$ make with the normal.

2A. Let $V = 2(x+1)^2(y+2)^2(z+3)^2\ volts$ in free space. At $P(2,-1,4)$, determine the following: ***(03)***

a) The electric potential

b) Magnitude of the electric field intensity

c) Volume charge density

2B. A circular disc of radius 'a' is uniformly charged with $\rho_s\ {}^{C}/_{m^2}$. If the disc lies on the $z = 0$ plane with its axis along the $z - axis$, prove that at point $(0,0,h)$: ***(03)***

$$\bar{E} = \frac{\rho_s}{2\varepsilon_0}\left\{1 - \frac{h}{(h^2 + a^2)^{1/2}}\right\} a_z V/m$$

2C. The capacitance of an air-filled parallel plate capacitor is 60 pF. When a dielectric slab whose thickness is half the distance between the plates is placed on one of the plates covering it entirely, the capacitance becomes 86 pF. Neglecting the fringing effects, determine the relative permittivity of the dielectric (upto 2 decimal places). ***(04)***

3A. Three equal positive charges of $8.01 \times 10^{-9}\ C$ each are located at the three corners of a square of side $0.4m$. From the fundamental analysis, determine the magnitude and direction of the electric field at the vacant corner point of the square. ***(03)***

3B. Two extensive homogeneous isotropic dielectrics meet in plane $z = 0$. Region 1: $z > 0$ contains the dielectric for which $\epsilon_{R1} = 4$, while region 2: $z < 0$ is characterized by $\epsilon_{R2} = 3$. In region 1, let $\bar{E}_1 = 5\boldsymbol{a_x} - 2\boldsymbol{a_y} + 3\boldsymbol{a_z}$ kV/m. Determine the angles both the field intensity vectors make with the interface. ***(03)***

3C. From the fundamentals, with a neat sketch, prove that, as per Poynting's theorem, the net power flowing out of a given volume is equal to the time rate of decrease in the energy stored within that volume minus the ohmic losses. ***(04)***

4A. With a neat diagram, explain and derive the expression for the induced electro motive force in a circuit that is in motion in a time varying magnetic field. ***(03)***

4B. A toroidal core has an average radius of 10 cm with a cross-sectional radius of 1 cm. If the core was made of steel ($\mu_R = 1000$) and the coil wound on it has 200 turns, calculate the amount of current that should flow to produce a magnetic flux of $0.5mWb$ in the core. ***(03)***

4C. In free space $(z \leq 0)$, a plane wave with $H_i = 10\cos(10^8 t - \beta z)\, a_x mA/m$ is incident normally on a lossless medium $(\varepsilon = 2\varepsilon_0, \mu = 8\mu_0)$ in region $z \geq 0$. Determine the expressions for the reflected wave H_r, E_r as well as the transmitted wave H_t, E_t. ***(04)***

5A. Let $\overline{E} = (1000a_x + 400a_z)e^{-j10y}\ V/m$ for a $250\ MHz$ uniform plane wave propagating in a perfect dielectric. If the maximum amplitude of the magnetic field intensity is $3\ A/m$, determine the following: ***(03)***

a) Relative permittivity of the dielectric
b) Relative permeability of the dielectric
c) $\bar{E}(x, y, z, t)$

5B. For Pyranol 1467 at 10^8 Hz, the conductivity $\sigma = 2.5 \times 10^{-3}\ \mho/m$, $\varepsilon_R = 4$ and $\sigma/\omega\varepsilon = 0.1$, determine the following: ***(03)***

a) Attenuation constant
b) Phase constant
c) Wavelength
d) Velocity of propagation of the uniform plane wave
e) Intrinsic impedance of the medium.

5C. With suitable diagrams, derive the expressions for reflection and transmission coefficients in plane wave shielding theory. Hence, prove that, the shielding effectiveness of an infinite sheet of good conductor is dependent on the reflection loss as well as absorption loss. ***(04)***